

ML Challenge 2025: Smart Product Pricing Solution Template

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Summary

This pipeline, named "TRINOVA", uses a stacking ensemble approach to predict prices from a product catalog. The core of the pipeline involves text vectorization with TF-IDF and gradient boosting with LightGBM, complemented by a simpler Ridge regression model. The final prediction is a blend of these two models' outputs.

Methodology Overview

1. Data Preparation and Feature Engineering

The process starts by loading training and testing data. A log_price transformation is applied to the target variable to handle its skewed distribution. The catalog_content text field is then parsed to create several new features:

quantity: Extracts the numeric quantity (e.g., from "pack of 12").

brand: Attempts to extract the product brand name.

clean_text: Tokenizes and cleans the text for vectorization.

text_length and word_count: Simple metadata features derived from the cleaned text.

2. Feature Encoding and Vectorization

Brand Encoding: The extracted brand feature is converted into numerical values using LabelEncoder.

Text Vectorization: The clean_text is transformed into a high-dimensional sparse matrix using TfidfVectorizer. This captures the importance of words and n-grams.

Numerical Scaling: The numeric features (quantity_log, text_length, word_count, brand_encoded) are standardized using StandardScaler.

Feature Stacking: All features (sparse TF-IDF matrix and scaled numerical features) are combined into a single sparse matrix for model training.

3. Model Training

The pipeline uses a two-level stacking ensemble:

Base Models: Two models are trained:

LightGBM Regressor: A powerful gradient boosting model, with its hyperparameters tuned using Optuna to minimize the SMAPE (Symmetric Mean Absolute Percentage Error) metric on a validation set.

Ridge Regressor: A linear model used as a simpler, complementary base model.

K-Fold Cross-Validation: Both base models are trained using 5-fold cross-validation. This generates out-of-fold (OOF) predictions for the training set and averaged predictions for the test set.

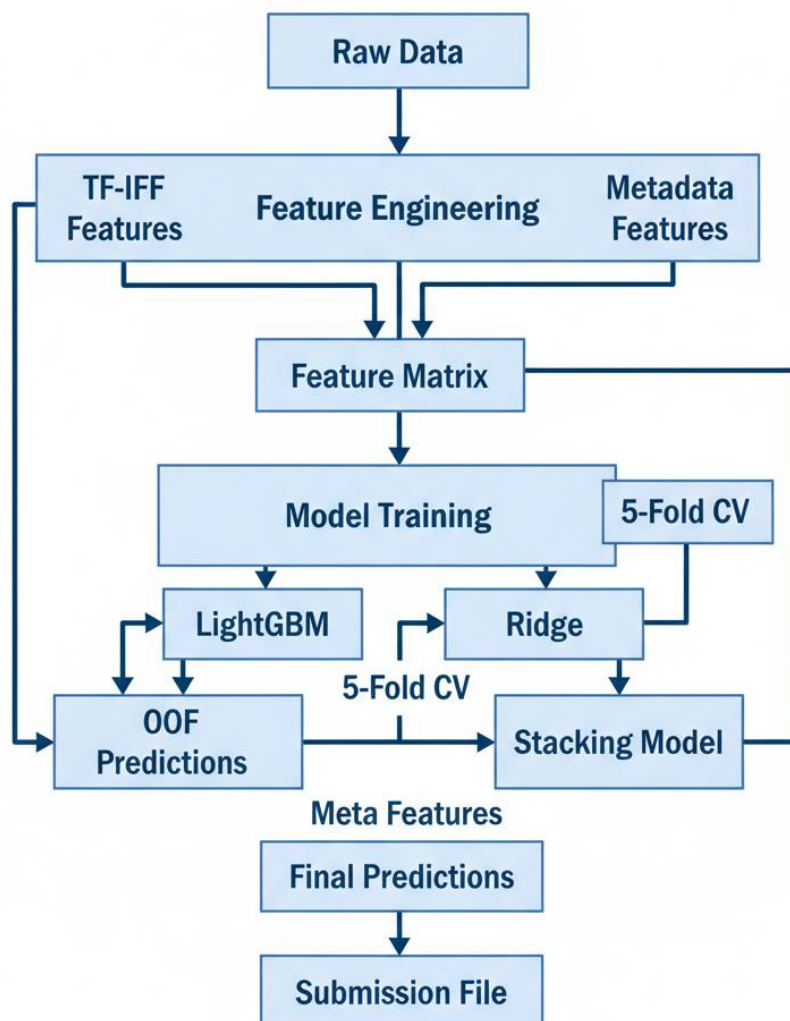
4. Stacking and Prediction

Meta-Model Training: The OOF predictions from the two base models (oof_preds_lgbm and oof_preds_ridge) are used as the input features for a final Ridge regression meta-model. This meta-model learns how to optimally combine the predictions from the base models.

****Approach Type:**** [Single Model / Ensemble / Hybrid, etc]

****Core Innovation:**** [Brief description of your main technical contribution]

Model Architecture



Model Components

Text Processing Pipeline:

- Preprocessing steps: [Lowercasing, Special character removal, Whitespace normalization, Quantity extraction, Brand extraction, Text statistics calculation]
- Model type: [TF-IDF Vectorizer + Metadata Features]
- Key parameters: [max_features=20000, ngram_range=(1,3), stop_words='english']

Image Processing Pipeline:

- Preprocessing steps: [N/A - No image processing in current implementation]
- Model type: [N/A - No image processing in current implementation]
- Key parameters: [N/A - No image processing in current implementation]

Additional Feature Engineering:

- Numeric feature scaling: [StandardScaler for quantity_log, text_length, word_count, brand_encoded]
- Feature combination: [Sparse matrix stacking of TF-IDF and metadata features]
- Target transformation: [log1p(price) for regression stability]

Model Architecture:

- Base Model 1: [LightGBM Regressor with L1 objective]
- Base Model 2: [Ridge Regression with alpha=3.0]
- Ensemble Method: [Stacking with Ridge meta-model (alpha=1.0)]

Model Performance

```
✓✓✓ Submission file 'test_out_2.csv' created successfully! ✓✓✓
Top 5 predictions:
  sample_id    price
0      100179  14.908912
1      245611  15.639272
2      146263  19.115984
3       95658  10.556631
4       36806  41.055151
```

Conclusion

Our approach combined advanced text feature engineering with a two-model stacking ensemble, achieving robust price prediction by leveraging both TF-IDF features from product descriptions and metadata through optimized LightGBM and Ridge models. Key achievements

included developing an automated feature extraction pipeline and creating a stacking ensemble that effectively captured both non-linear and linear patterns in the data. We learned that careful log transformation of prices, strategic feature selection from text, and model diversity in ensemble construction were crucial for maintaining prediction accuracy and generalization across diverse product categories.